Konn3ct Networking Evolution: Before vs. Now

This document highlights the evolution of the Konn3ct network architecture, transitioning from our previous infrastructure to the newly engineered, modern architecture.

WebRTC UDP Port and Firewall Differences

The most critical difference for network engineers is the shift in required firewall rules and UDP port ranges for media traffic.

Firewall Configuration: Before

It required a massive range of UDP ports because of the separation between audio (FreeSWITCH) and video (Kurento) engines.

Protocol	Port Range	Purpose
TCP	80, 443	Standard HTTP/HTTPS web traffic (Nginx)
UDP	**16384 - 32768**	Massive port range for WebRTC Audio & Video streams.
TCP / UDP	3478, 5349	Coturn (STUN/TURN) Hosted on the same server.

Firewall Configuration: Now

The new architecture is much more deliberate and conservative with port usage, reducing the attack surface and making firewall rules easier to audit.

Protocol	Port Range	Purpose
TCP	80, 443	HTTP/HTTPS pointing to the Cloud Load Balancer (not directly to the server).
UDP	**10000 - 20000**	Narrower, dedicated port range specifically for Mediasoup WebRTC routing directly to the K8s Worker Nodes.
TCP	3000	Internal K8s health checks and Ingress routing (restricted to internal VPC).

Key Firewall Difference: Instead of opening `16384-32768` (16,384 ports) directly to a single monolith, you now open `10000-20000` (10,000 ports) on your **Kubernetes Worker Node Security Groups**, while web traffic (TCP 80/443) is safely offloaded to a Managed Cloud Load Balancer.

Tighter Security: The UDP media port range has been reduced by nearly 40%, and edge web traffic is now shielded by managed Cloud Load Balancers rather than directly exposing the application server.